TerraMind on SmallMinesDS: Experiment Report

Oct 21, 2026 · Asım Emre Arslan

Summary

We trained a shallow UNet decoder on SmallMinesDS 2022 (Sentinel-1 + Sentinel-2, 128 × 128 px) on top of TerraMind v1 tiny, either pretrained and frozen or randomly initialised and trainable, with and without D4 geometric augmentation, for N = 1 to 250 training images and 5 seeds (160 runs; 50 epochs, AdamW, cosine learning rate 5e-3 → 1e-5). The frozen pretrained encoder gives the higher test IoU at every N ≥ 10 in all seeds, and the gap grows with data: 0.707 vs 0.584 at N = 250 with augmentation. With 10 images it already matches what the trainable encoder reaches with 250. Augmentation adds +0.02 to +0.04 IoU for the frozen encoder at every N ≥ 3 and has no consistent effect on the trainable one.

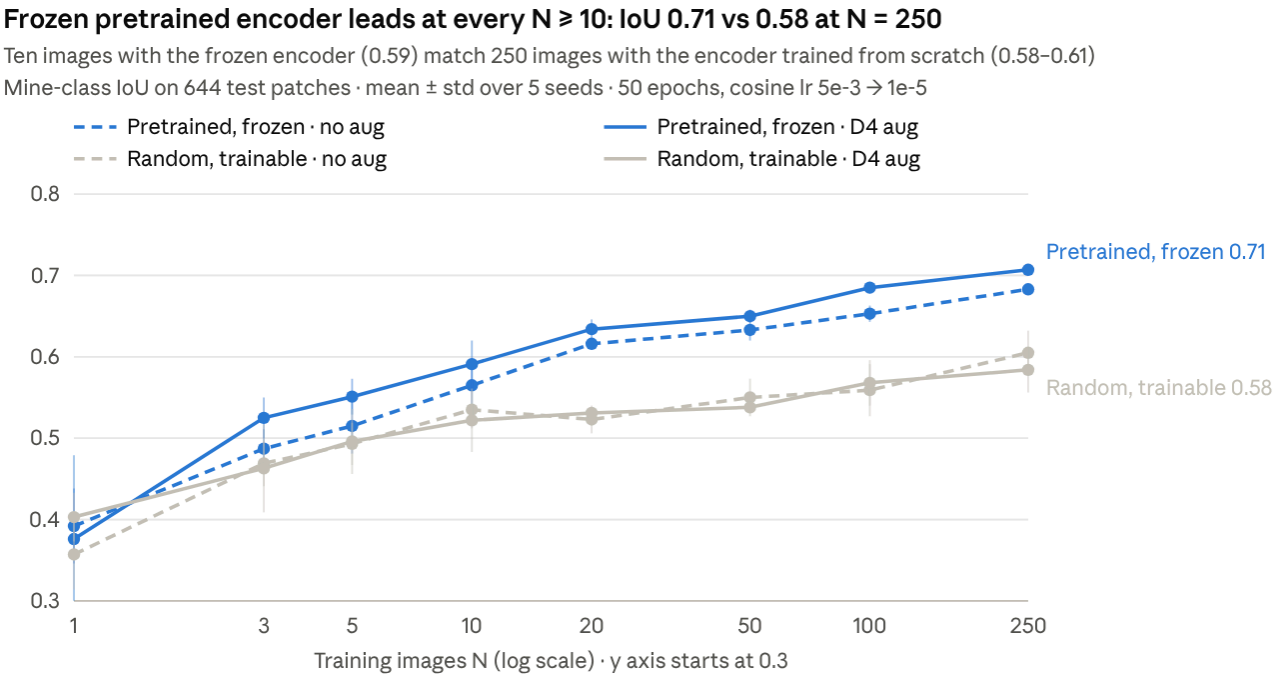


Figure 1. Mine-class IoU on the test set against the number of training images, mean ± std over 5 seeds

Table 1. Mine-class IoU on the test set (mean ± std over 5 seeds).

N	Pretrained, frozen	Pretrained, frozen + aug	Random init, trainable	Random init, trainable + aug
1	0.392 ± 0.046	0.376 ± 0.103	0.357 ± 0.043	0.403 ± 0.030

N	Pretrained, frozen	Pretrained, frozen + aug	Random init, trainable	Random init, trainable + aug
3	0.487 ± 0.024	0.525 ± 0.025	0.469 ± 0.028	0.463 ± 0.054
5	0.515 ± 0.034	0.551 ± 0.022	0.493 ± 0.026	0.496 ± 0.040
10	0.565 ± 0.023	0.591 ± 0.029	0.535 ± 0.023	0.522 ± 0.039
20	0.616 ± 0.007	0.634 ± 0.012	0.523 ± 0.017	0.531 ± 0.007
50	0.633 ± 0.013	0.650 ± 0.008	0.550 ± 0.023	0.538 ± 0.005
100	0.653 ± 0.010	0.685 ± 0.007	0.559 ± 0.032	0.568 ± 0.028
250	0.683 ± 0.004	0.707 ± 0.002	0.605 ± 0.027	0.584 ± 0.028

Table 2. IoU gain from D4 augmentation (augmented minus unaugmented run with the same seed and training images; mean ± std over 5 seeds, in brackets the seeds out of 5 that improved).

N	Pretrained, frozen	Random init, trainable
1	−0.016 ± 0.073 (3/5)	+0.046 ± 0.044 (4/5)
3	+0.038 ± 0.016 (5/5)	−0.006 ± 0.045 (2/5)
5	+0.035 ± 0.027 (5/5)	+0.003 ± 0.034 (3/5)
10	+0.027 ± 0.010 (5/5)	−0.013 ± 0.024 (2/5)
20	+0.018 ± 0.007 (5/5)	+0.008 ± 0.017 (3/5)
50	+0.016 ± 0.005 (5/5)	−0.011 ± 0.024 (2/5)
100	+0.031 ± 0.007 (5/5)	+0.010 ± 0.007 (5/5)
250	+0.024 ± 0.003 (5/5)	−0.021 ± 0.034 (2/5)

Code, per-run results and full details: github.com/asimarslan11/SmallMinesDS-Experiment-Project.